

Department Of CSE(AI&ML).

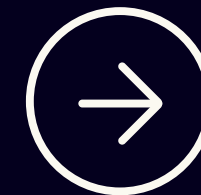
PROJECT PRESENTATION

Presented by the Team Path Guiders

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E-GUIDE

An E-portal to guide the path of placement to students in the field of Engineering and predicting the possibility percentage of their placement.



Project vision and mission

The purpose of this project is to use data-driven insights to guide career planning and institutional strategy in the domain of campus recruitment.

To be the leading predictive tool in campus recruitment, empowering students to proactively enhance their skillsets and academic performance, and enabling educational institutions to optimize their placement strategies for maximum student success.

01.

Prediction Accuracy: To accurately forecast student placement status (Placed/Not Placed) and estimate potential salary using highly reliable Random Forest classifiers.

02.

Student Empowerment: To provide students with personalized, actionable feedback based on their academic and skill metrics (CGPA, internships, hackathons, skills, etc.) so they can focus their efforts where they matter most.

03.

Strategic Insight: To offer educational institutions clear insights into the key features influencing placement outcomes (Feature Importance) to help them refine curriculum, workshops, and career readiness programs.



Features

1. Dual Machine Learning Prediction 📊

The system utilizes two distinct Random Forest models to provide comprehensive career insights:

- Placement Status: A classification model predicts the likelihood of the student being placed.
- Expected Salary: A regression model estimates the potential salary for students who are predicted to be successful.

2. Strict Confidence Threshold for Placement 🎯

The core prediction logic uses a high confidence score threshold (e.g., 90%) to determine a "High Chance" placement outcome. This feature ensures that:

- The system requires the model to be highly certain before classifying a student as 'Placed', increasing the precision of the positive predictions.
- This stricter standard helps provide more realistic guidance, distinguishing highly confident results from marginal ones.

3. Comprehensive Feature-Based Input & Analysis 📝

Predictions are based on a detailed input profile covering essential student attributes:

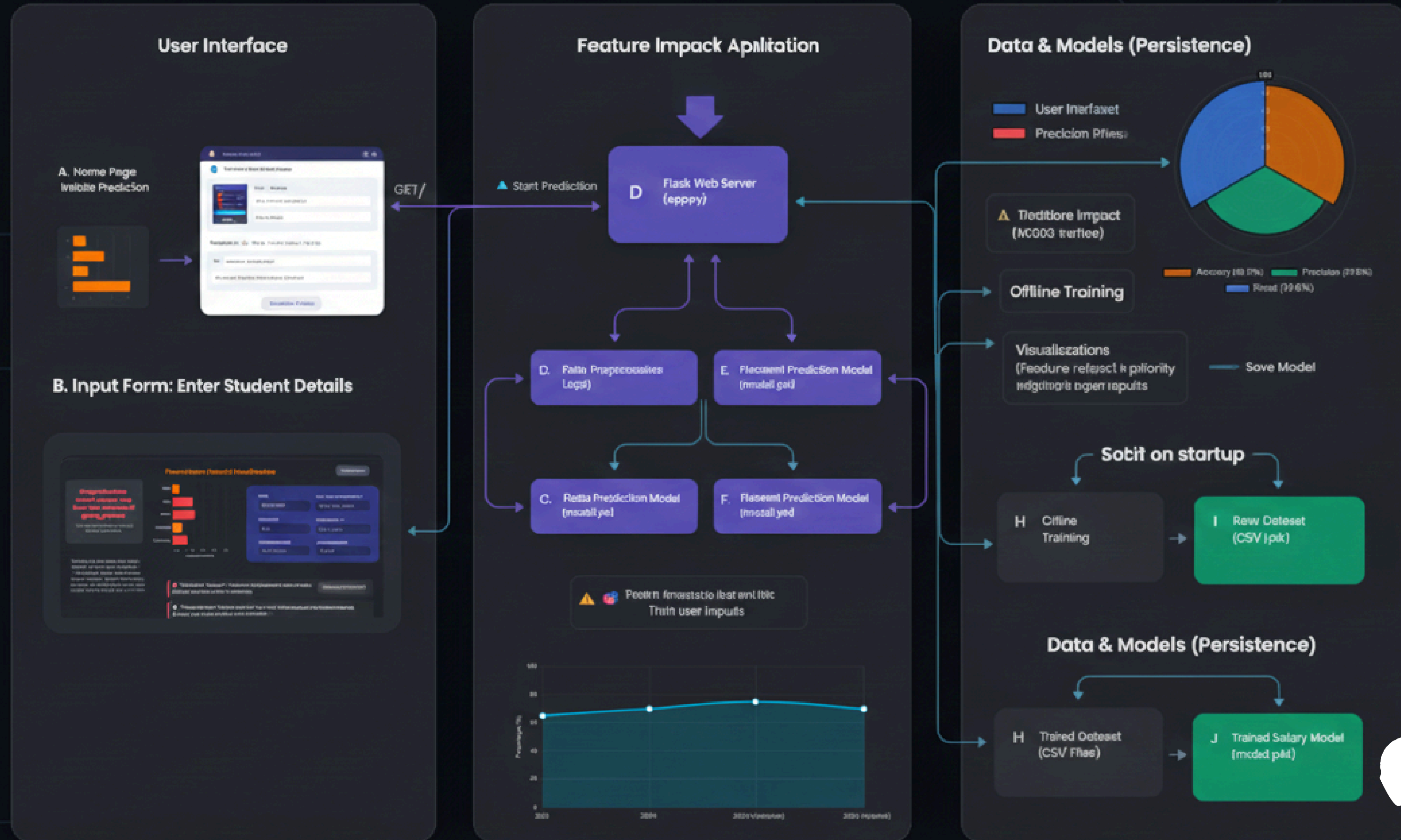
- Academic Performance: Includes CGPA, and final year percentages (TW and TE).
- Skill Development: Accounts for participation in projects, workshops, mini-projects, internships, and hackathons, alongside communication and technical skills.

3. Dedicated Placement Query Chatbot & Guidance 💬

The project integrates an interactive Chatbot specifically designed for career and placement guidance.

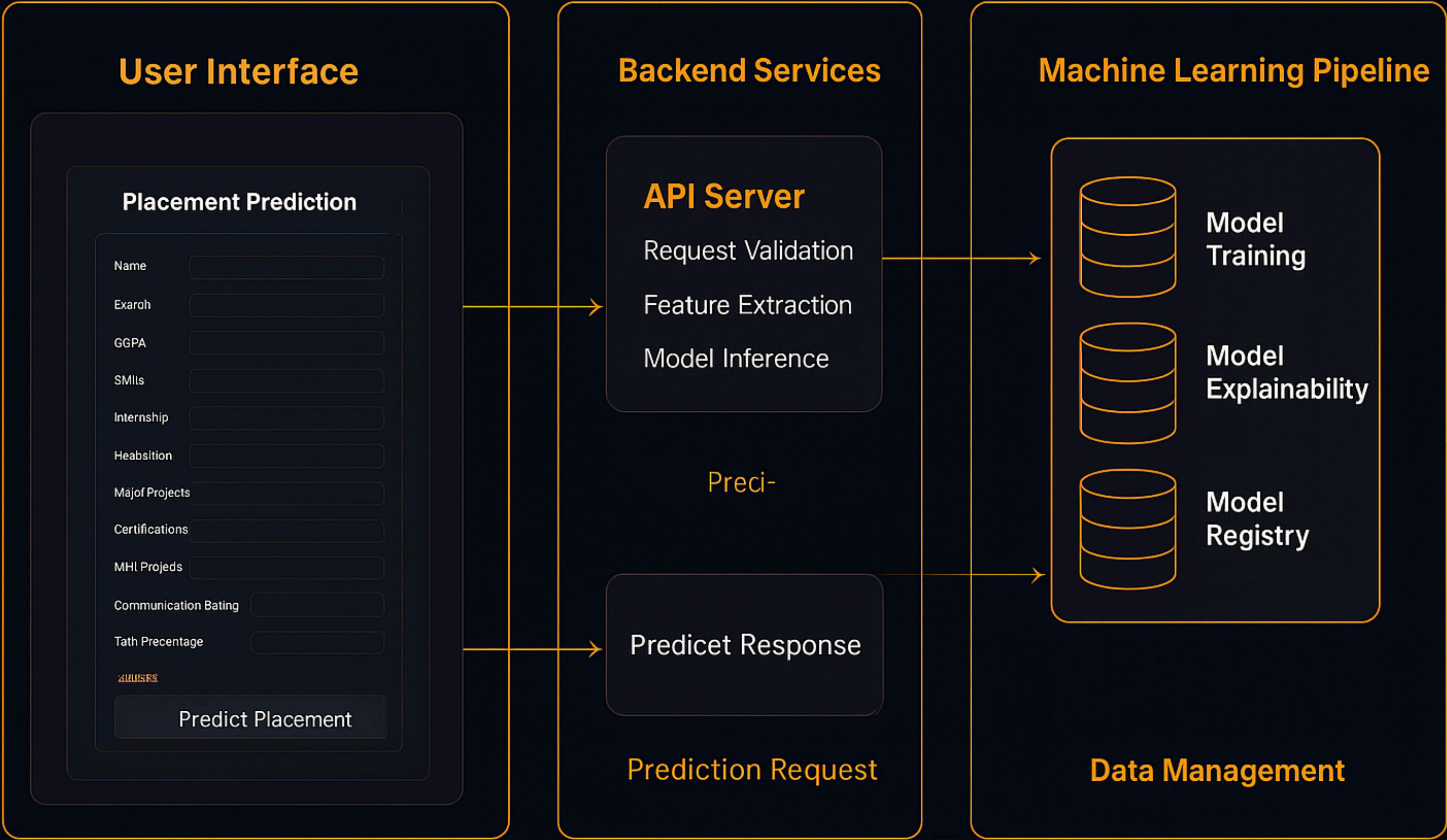
- Skill Gap Analysis: The chatbot accepts user queries regarding skills and provides suggestions for programming languages (e.g., C, C++, Python) and technologies critical for placement success.
- Feature Impact Explanation: It offers interactive assistance to explain how key features like Internships, Hackathons, and CGPA affect their overall placement probability, enabling students to make informed decisions.
- Personalized Advice: Delivers tailored feedback on areas to improve, complementing the quantitative prediction with actionable guidance.





MODEL

Placement Prediction — System Architecture



INITIATE PLACEMENT PREDICTION PROTOCOL

A Placement Prediction App is a tool designed to leverage cutting-edge algorithms and synthesized data streams to provide real-time, actionable insights into your potential career trajectory. Analyze, optimize, and secure your future with precision. Your success is the only output that matters.

EXECUTE PREDICT

Jeet – Your Placement Success Assistant

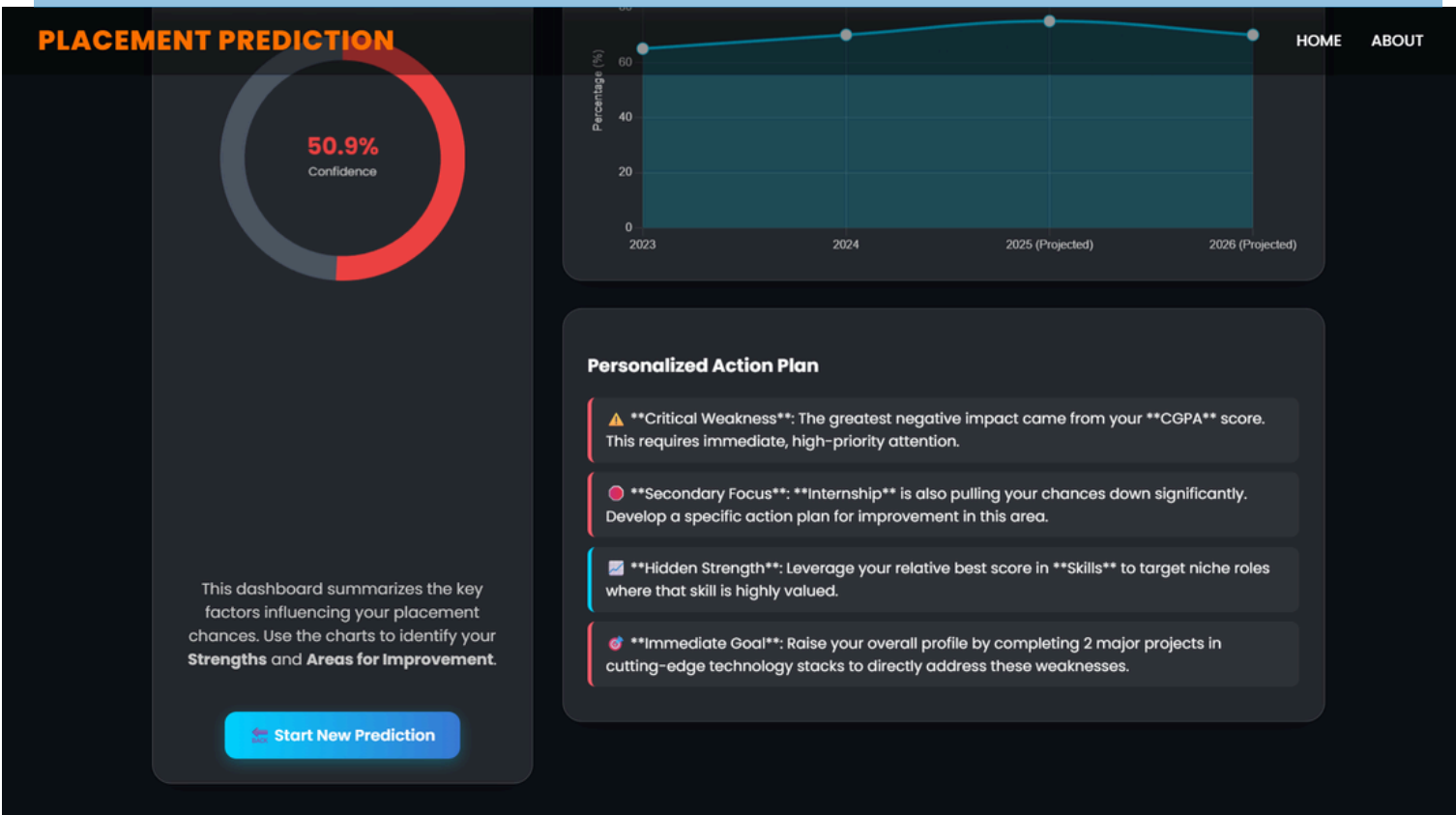
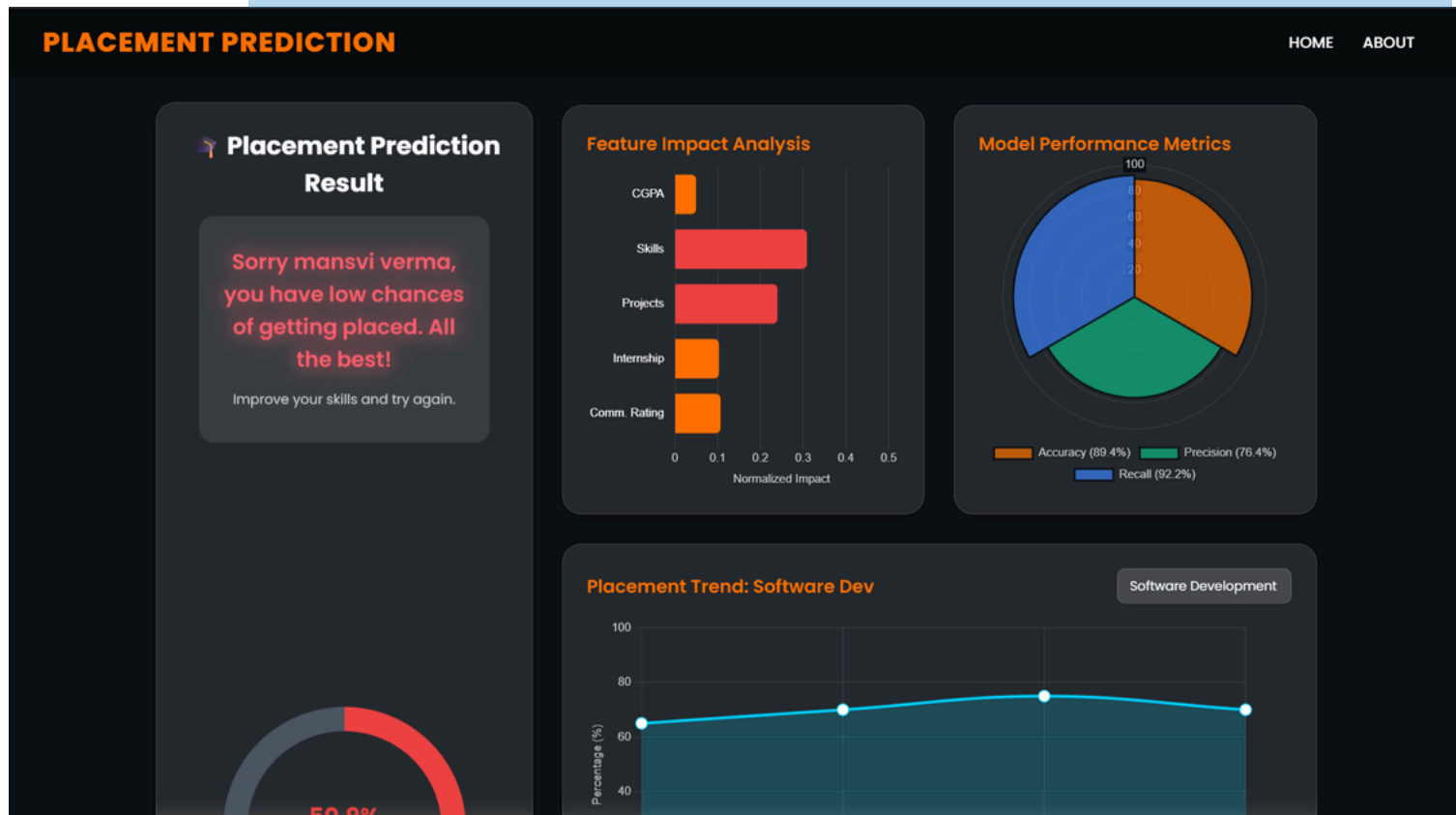
(Jeet means 'Win' in placement!)



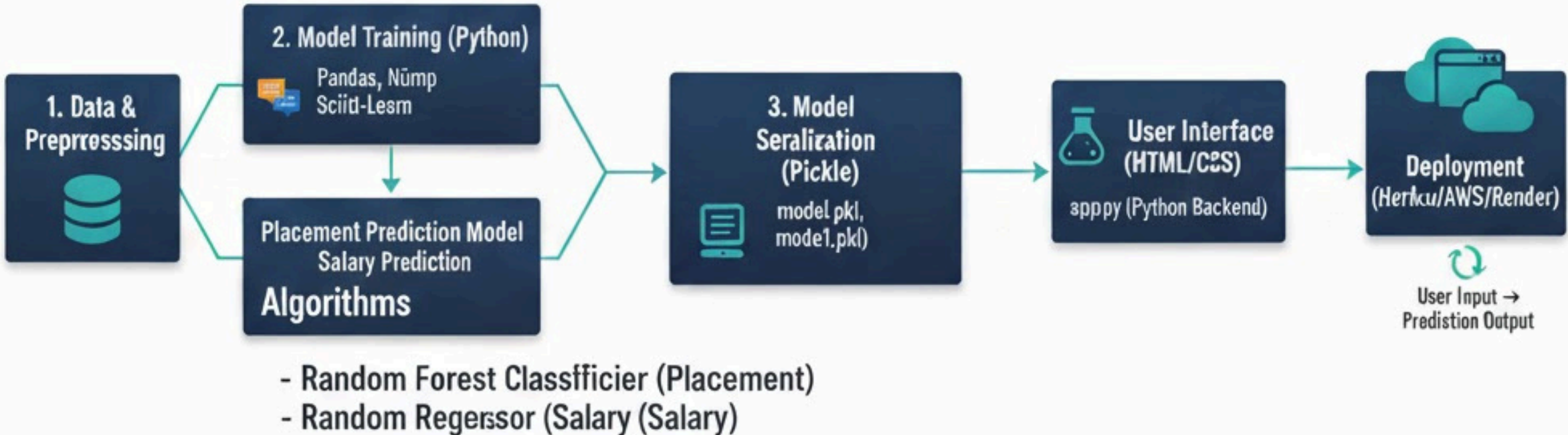
Namaste! I am Jeet, your personal Placement Assistant. I am here to help you ****win**** your placement by answering any queries you have about required skills, CGPA, interview tips, or projects. Ask me anything to get started!

Meet Jeet, your personalized chatbot to guide you for your jeet in placements

THIS IS HOW OUR GUIDE LOOKS



TECH STACK & APPLICATION WORKFLOW



Core Language: Python (for all data manipulation and model building).

Machine Learning Framework: Scikit-Learn (for model training and evaluation).

Data Handling: Pandas and NumPy (for efficient data preprocessing).

Deployment: Flask (a lightweight Python web framework for hosting the prediction service).

Model Persistence: Pickle (used to save and load the trained models for quick prediction).



THANK YOU